

[I225] Statistical Signal Processing(E) Office Hour 1

1. Consider a discrete-time LTI system used to create a simple echo effect. The system has the following impulse response

$$h[n] = \delta[n] + 0.4\delta[n - 2]$$

The input signal is:

$$x[n] = \{1, 2, 0, -1\}$$

where the first element corresponds to $n = 0$.

- (a) Explain what the impulse response means physically.
- (b) Calculate the output signal $y[n] = x[n] * h[n]$.

2. Consider a continuous-time linear time-invariant system with impulse response:

$$h(t) = e^{-2t}U(t)$$

The input random process is:

$$X(t) = f(t) + v(t), \quad f(t) = e^{-t}U(t)$$

where $v(t)$ is a zero-mean random process, so $E\{v(t)\} = 0$.

- (a) Find the mean of the input process $\eta_X(t)$.
 - (b) Find the mean of the output process $\eta_Y(t)$.
3. Consider a zero-mean white noise process $W(t)$ injected into an LTI system at $t = 0$. The input autocorrelation is:

$$R_{WW}(\tau) = q\delta(\tau)$$

The impulse response of the system is:

$$h(t) = e^{-ct}U(t), \quad c > 0$$

- (a) Find the output power $E\{Y^2(t)\}$.
 - (b) Find the steady-state output power as $t \rightarrow \infty$.
4. Suppose $X(t)$ is a wide-sense stationary random process with autocorrelation function:

$$R_{XX}(\tau) = \sigma^2 e^{-a|\tau|}, \quad a > 0$$

- (a) Find the power spectral density $S_{XX}(\omega)$.
- (b) Verify the signal power from the autocorrelation function.

5. Consider a zero-mean white noise process $X(t)$ with power spectral density:

$$S_{XX}(\omega) = \frac{N_0}{2}$$

The process passes through an ideal low-pass LTI system with frequency response:

$$H(j\omega) = \begin{cases} 1, & |\omega| \leq \omega_c, \\ 0, & |\omega| > \omega_c. \end{cases}$$

(a) Find the output power spectral density $S_{YY}(\omega)$.

(b) Find the total output power.